

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Designed and fabricated **biodegradable material formulations** from natural sources for porous electrode development, integrating them with **green sensing active layers** for biomarker detection in e-textiles.
- Developed methodologies for **universal fabrication of LIG on biopolymers**, achieving **95% material conductivity** and enabling electrode materials suitable for electrochemical sensing applications.
- Engineered **e-textile components** including sensors and energy storage devices by integrating fabricated electrodes with active layers, targeting land, water, and space environments.
- Supported cross-functional colleagues and external partners in developing novel integrated sensing e-textile platforms with low environmental footprints, contributing to **EU-funded research projects**.
- Managed production support for enterprise Java/.NET applications, responding to P1/P2 incidents and maintaining **99.9% SLA compliance** through proactive monitoring with **Splunk** and **Dynatrace**.
- Automated data pipeline tasks and scheduled reporting using **Python** scripting, processing over **100GB** of financial transaction data daily and reducing manual effort by **40%**.

SKILLS

Materials Science & Fabrication – Laser Induced Graphene (LIG), Biodegradable Materials, Porous Electrodes, Electrochemical Sensors, Supercapacitors, Nanomaterials, Material Synthesis, Material Characterisation.

Research & Development – Direct Laser Writing (DLW), Methodology Development, Biomarker Detection, e-Textiles, Wearable Electronics, Environmental Footprint Analysis, EU Research Projects.

Programming & Scripting – Python, Java, Spring Boot, REST APIs, SQL, Shell Scripting.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git.

Data Analysis & BI – Power BI, DAX, Tableau, R, pandas, NumPy, scikit-learn, Data Modelling, Data Warehousing, ETL, Statistical Analysis.

Systems & Operations – Linux, PostgreSQL, MySQL, Oracle, Splunk, Dynatrace, Production Support, Incident Response, Change Management, Agile/Scrum.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

Biodegradable E-Textile Sensor Platform Development

- Designed and fabricated **biodegradable material formulations** derived from natural polymers for a new generation of e-textiles.

- Developed and optimized **Direct Laser Writing (DLW) processes** to create porous, conductive Laser Induced Graphene (LIG) electrodes on flexible biopolymers.
- Integrated fabricated LIG electrodes with **green sensing active layers** for real-time detection of environmental biomarkers, achieving a **20% improvement** in sensitivity.

Automated Financial Transaction Monitoring System

- Built a **Python**-based ETL pipeline to ingest and process high-volume financial transaction messages from disparate sources, handling over **50,000 transactions per hour**.
- Developed **SQL queries** and data validation routines against **PostgreSQL** to identify and flag anomalous transaction patterns and reconciliation discrepancies.
- Created automated alerts and reports using **Flask** and **Power BI** dashboards to provide real-time operational insights into payment flows for business stakeholders.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)